

Obstacle-Aware Locomotion for Humanoid Robots: A Comparative Study of PPO and TD3 with LiDAR Perception

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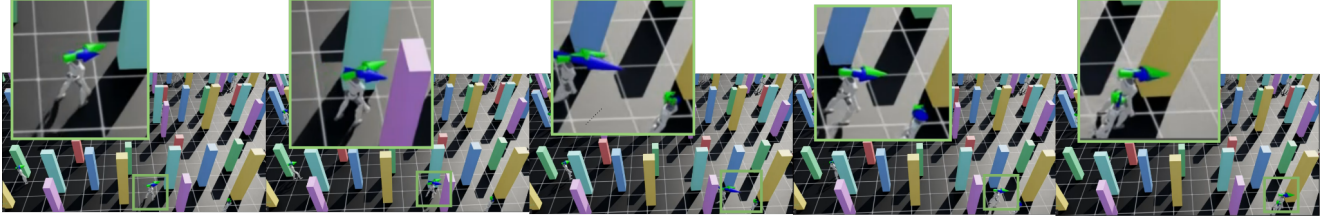


Figure 1. Test time inference episode under the PPO policy. The humanoid successfully avoids collision courses while navigating through the obstacle world. The blue arrow indicates the action direction, while the green arrow provides the reference direction.

Abstract

We study the problem of obstacle aware locomotion for the Unitree G1 29-DoF humanoid robot using deep reinforcement learning in simulation. We formulate two distinct Markov Decision Processes: an empty world for baseline locomotion and an obstacle world that introduces randomly placed cuboid obstacles with collision termination and proximity penalties. For each world we train two families of agents one relying solely on proprioceptive observations and one additionally receiving a front facing LiDAR point cloud using both Proximal Policy Optimization (PPO) and Twin Delayed Deep Deterministic Policy Gradient (TD3). Through systematic ablations we find: (i) PPO outperforms TD3 in both the environment and with and without lidar ablation; (ii) LiDAR perception raises PPO’s obstacle world timeout/non collision rate from near-zero to 94.5%, but TD3 cannot leverage LiDAR at all episode lengths actually decrease relative to TD3 without LiDAR, revealing that TD3 suffers with very delicate parameter tuning and training is difficult for 1,720 dimensional observations; and (iii) two stage pre training in empty world first followed by obstacle world fine tuning is critical, yielding $34\times$ larger episode lengths than the baseline empty world policy. All experiments are conducted in NVIDIA Isaac Lab using 4,096 parallel environments on a single GPU.

Introduction

Bipedal locomotion in cluttered environments remains a central challenge in robotics. While model based controllers excel in structured settings, they degrade gracefully under sensor noise and unmodelled contacts. Reinforcement learning (RL) particularly when coupled with massively parallel physics simulation has recently produced locomotion controllers that match or exceed model-based alternatives on rough terrain [Lee et al. 2020, Kumar et al. 2021, Margolis et al. 2022]. However, most prior work focuses on terrain adaptation and ignores dynamic or static obstacles that require active avoidance.

The Unitree G1, with its 29 degrees of freedom and human proportioned morphology, presents a rich testbed for obstacle aware locomotion research. We make the following contributions:

- A sim ready MDP formulation for a 29 DoF humanoid in both free and obstacle cluttered environments, implemented in Isaac Lab.
- A systematic ablation comparing on policy (PPO) and off policy (TD3) algorithms across two observation modalities (proprioception only vs. proprioception with exteroceptive LiDAR).
- An empirical analysis of curriculum free transfer from empty world to obstacle world policies.
- All code, configurations, and training scripts are available at https://github.com/mahajanparth/G1_Humanoid-Obstacle_Avoidance_RL.

Related Work

Legged locomotion via RL. Lee et al. [2020] demonstrated that a teacher student framework can transfer rough terrain locomotion policies from simulation to real quadrupeds. Kumar et al. [2021] introduced Rapid Motor Adaptation (RMA), learning an online adaptation module conditioned on extrinsics. For bipeds, Radosavovic et al. [2024] and concurrent work show sim to real transfer of whole body locomotion on the Unitree H1 platform.

Obstacle avoidance for legged robots. Navigation aware locomotion has been studied by coupling high level planners with low level locomotion policies [Hoeller et al. 2024, Zhuang et al. 2023]. Our work differs in that we embed obstacle awareness directly into the locomotion policy via LiDAR observations rather than relying on a separate planning layer.

On-policy vs. off-policy RL for continuous control. PPO [Schulman et al. 2017] dominates locomotion benchmarks

due to sample efficiency in the massively parallel setting. TD3 [Fujimoto et al. 2018] has demonstrated strong results in lower dimensional MuJoCo tasks but is less studied at humanoid scale. We wanted to first hand see the direct comparison of the two on a 29 DOF humanoid.

Background

Markov Decision Process

We formulate each task as a finite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, r, \gamma, \rho_0)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $\mathcal{T}$ the (unknown) transition dynamics, r the reward function, $\gamma \in [0, 1)$ the discount factor, and ρ_0 the initial state distribution. The agent observes a partial observation $o_t \in \mathcal{O}$ and learns a policy $\pi_\theta : \mathcal{O} \rightarrow \mathcal{A}$.

Proximal Policy Optimization (PPO)

PPO [Schulman et al. 2017] is an on policy actor critic algorithm that constrains policy updates via a clipped surrogate objective:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (1)$$

where $r_t(\theta) = \pi_\theta(a_t|o_t) / \pi_{\theta_{\text{old}}}(a_t|o_t)$ is the probability ratio and $\hat{A}_t$ is the Generalised Advantage Estimate (GAE) [Schulman et al. 2016]. The full objective adds a value-function loss and an entropy bonus:

$$\mathcal{L}(\theta) = \mathcal{L}^{\text{CLIP}}(\theta) - c_1 \mathcal{L}^{\text{VF}}(\theta) + c_2 \mathcal{H}[\pi_\theta]. \quad (2)$$

PPO is particularly well suited to massively parallel simulation because rollouts from all environments can be batched into a single update step.

Twin Delayed Deep Deterministic Policy Gradient (TD3)

TD3 [Fujimoto et al. 2018] is an off policy actor critic algorithm that extends DDPG [Lillicrap et al. 2016] with three key improvements: (i) *clipped double-Q learning* to reduce overestimation bias, (ii) *delayed policy updates* (the actor is updated every d critic steps), and (iii) *target policy smoothing* via noise injected into the target action:

$$y = r + \gamma \min_{i=1,2} Q_{\theta'_i}(s', \pi_{\phi'}(s') + \epsilon), \quad (3)$$

$$\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c).$$

$$\mathcal{L}(\theta_i) = \mathbb{E}[(y - Q_{\theta_i}(s, a))^2]. \quad (4)$$

TD3 learns from a replay buffer $\mathcal{D}$ and is therefore sample efficient for long horizon tasks, though it requires careful tuning of exploration noise.

MDP Formulation

Two-World Curriculum

We define two distinct environments:



Figure 2: The two task MDPs used in this work. Left: the empty-world MDP used for baseline locomotion pretraining. Right: the obstacle-world MDP with randomly placed cuboid obstacles used for obstacle-aware locomotion.

Empty World ($\mathcal{E}$). A flat infinite ground plane with no obstacles. The robot is rewarded purely for velocity tracking and locomotion quality. This world is used to establish a locomotion baseline and pretrain policies that are later fine-tuned.

Obstacle World ($\mathcal{W}$). The same ground plane augmented with up to six kinematic cuboid obstacles, randomly placed at each episode reset. Obstacle positions, counts, and the corridor in front of the robot are all randomised to prevent overfitting to a fixed layout.

Robot Platform

We use the **Unitree G1** humanoid robot with **29 actuated joints**: 12 for the legs, 7 per arm, and 3 for the waist. The robot stands approximately 1.3 m tall with a target standing height of 0.78 m for the base link. All joints are controlled via *position targets* with a PD controller running at 200 Hz; the RL policy runs at 50 Hz (decimation factor 4).

Observation Space

We study two observation modalities.

Without LiDAR (proprioceptive only). The policy observation o_t at each step contains:

Component	Dim	Scale / Noise
Base angular velocity	3	$\times 0.2, \sigma=0.2$
Projected gravity vector	3	$\sigma=0.05$
Velocity commands (v_x, v_y, ω_z)	3	—
Joint positions (rel. default)	29	$\sigma=0.01$
Joint velocities	29	$\times 0.05, \sigma=1.5$
Last action	29	—
Per-step total	96	
With history ($T=5$)	480	

The critic additionally receives the base linear velocity (3D), giving a 483-dimensional critic observation per step. Gaussian observation noise is applied during training only to improve robustness.

With LiDAR. A `MultiMeshRayCaster` sensor is mounted on the torso link at offset (0.2, 0.0, 0.15) m and aligned to the robot’s base frame. It fires 8 channels $\times$ 31 rays = 248 rays over a 180° horizontal and 30° vertical field of view at 6° resolution with a maximum range of 20 m. LiDAR ranges are appended to both policy and critic observations, increasing the dimension by $248 \times 5 = 1,240$.

per history window. An explicit normalization is applied to the LiDAR ; also empirical normalization inside the PPO runner handles scaling automatically.

Action Space

The action $a_t \in \mathbb{R}^{29}$ represents joint position offsets from the robot’s default standing pose, scaled by 0.25 radians. A low-level PD controller converts these targets to joint torques within the simulator.

Reward Functions

Empty World Rewards The empty-world reward $r_t^{\mathcal{E}}$ consists of the following 19 terms:

Term	Description	Weight
<i>Tracking</i>		
track_lin_vel_xy	Exp. tracking of commanded v_x, v_y	+1.0
track_ang_vel_z	Exp. tracking of commanded ω_z	+0.5
<i>Survival</i>		
alive	Constant bonus per step	+0.15
<i>Smoothness & Efficiency</i>		
base_linear_velocity	Penalise z-axis linear velocity	-2.0
base_angular_velocity	Penalise xy angular velocity	-0.05
joint_vel	L2 joint velocity	-0.001
joint_acc	L2 joint acceleration	-2.5×10^{-7}
action_rate	Penalise action change rate	-0.05
energy	Penalise $\tau \cdot \dot{q}$	-2×10^{-5}
<i>Posture</i>		
flat_orientation_l2	Penalise body tilt	-5.0
base_height	Penalise deviation from 0.78 m	-10.0
dof_pos_limits	Joint limit violation	-5.0
joint_deviation_arms	Arm joints near default	-0.1
joint_deviation_waists	Waist joints near default	-1.0
joint_deviation_legs	Hip roll/yaw near default	-1.0
<i>Gait Quality</i>		
gait	Phase-synchronised footstep reward	+0.5
feet_clearance	Swing-foot height clearance	+1.0
feet_slide	Penalise foot sliding	-0.2
undesired_contacts	Penalise non-foot contacts	-1.0

The velocity-tracking terms use the exponential kernel $r = \exp(-\|e\|^2/\sigma^2)$ with $\sigma^2 = 0.25$. The gait reward enforces a walking pattern with period $T_g = 0.8$ s and phase offset $\Delta\phi = 0.5$ (alternating feet), using a stance threshold of 0.55.

Obstacle World Rewards The obstacle-world reward $r_t^{\mathcal{W}}$ extends $r_t^{\mathcal{E}}$ with two additional terms:

Term	Description	Weight
object_hit	Torso contact force with obstacles	-2.0
lidar_proximity	Rays within $d_{\text{safe}}=0.8$ m of obstacle	-0.5

The `object_hit` penalty is computed from the filtered contact force matrix $\|\mathbf{F}_{\text{mat}}\|$ (obstacles only, via `filter_prim_paths_expr`), ensuring ground contacts do not contribute. The `lidar_proximity` penalty sums $\max(0, d_{\text{safe}} - d_i)/d_{\text{safe}}$ over all 248 rays, providing a smooth keep-away signal before physical contact occurs.

Termination Conditions

Condition	Threshold
time_out	Episode length > 20 s
base_height	Root height < 0.2 m
bad_orientation	Body tilt > 0.8 rad
obstacle_collision*	Contact force with obstacle > 1 N

*Obstacle world only.

Algorithms and Hyperparameters

PPO Configuration

We use an actor critic architecture. The actor receives the policy observation group; the critic receives a privileged observation group that additionally includes the ground-truth base linear velocity.

Hyperparameter	Value
<i>Network architecture (both actor and critic)</i>	
Hidden layers	[1024, 1024, 512, 256, 128]
Activation	ELU
Output distribution	Gaussian, $\sigma_0 = 1.0$
<i>Rollout & update</i>	
Steps per env per update	24
Number of envs	4,096
Mini-batches	4
Learning epochs per update	5
Total rollout steps (batch)	98,304
<i>Optimisation</i>	
Learning rate	1×10^{-3} (adaptive)
Clip parameter ϵ	0.2
Entropy coefficient c_2	0.01
Value loss coefficient c_1	1.0
Discount factor γ	0.99
GAE parameter λ	0.95
Desired KL divergence	0.01
Max gradient norm	1.0
Empirical normalization	True
<i>Training</i>	
Max iterations	50,000
Best checkpoint criterion	Max mean episode return

The adaptive learning rate schedule adjusts α up or down by $1.5\times$ based on whether the measured KL divergence exceeds the desired threshold, stabilising training under varying reward scales.

TD3 Configuration

We use TD3 with the same MLP architecture as PPO to ensure a fair comparison of the algorithmic components. To stabilise training at 29 DoF humanoid scale we apply several modifications over the standard TD3 baseline: (i) a lower actor learning rate ($\alpha_{\text{actor}} = 0.1 \times \alpha_{\text{critic}}$) to reduce policy oscillation, (ii) a delayed policy update ratio of $d = 8$ to allow the critic to stabilise before the actor is updated, (iii) norm-based gradient clipping ($\|\nabla\| \leq 1.0$), and (iv) near-zero initialisation of the actor’s output layer to encourage small initial actions.

Hyperparameter	Value
<i>Network architecture (policy & Q-networks)</i>	
Hidden layers	[1024, 1024, 512, 256, 128]
Activation	ELU
Twin Q-networks	Yes
<i>Replay buffer & sampling</i>	
Replay buffer size	1,000,000
Batch size	256
Learning starts	10,000 steps
Gradient steps per env step	4
<i>Optimisation</i>	
Critic learning rate	1×10^{-4}
Actor learning rate	1×10^{-5} ($0.1 \times$ critic)
Discount factor γ	0.99
Soft update coefficient τ	0.005
Policy delay d	8
Target policy noise σ	0.2
Noise clip c	0.5
Max gradient norm	1.0
<i>Training</i>	
Total timesteps	5,000,000
Seed	42

Experimental Setup

Simulation Framework

All experiments are conducted in **NVIDIA Isaac Lab** [Mittal et al. 2023], a modular framework built on top of Isaac Sim (PhysX 5) that provides GPU-accelerated parallel simulation. We run 4,096 parallel environments on a single NVIDIA GPU, achieving approximately 50,000 environment steps per second for proprioceptive policies.

Simulation Parameter	Value
Physics engine	PhysX 5 (GPU)
Physics timestep Δt_{sim}	5 ms (200 Hz)
Control decimation	$4 \times$ (50 Hz)
Number of training environments	4,096
Environment spacing (empty world)	2.5 m
Environment spacing (obstacle world)	6.0 m
Ground plane	Infinite (plane), $\mu_s = 1.0$
<i>Obstacle world specific</i>	
Obstacle count per env	6 cuboids (0–5 active per reset)
Obstacle dimensions	$(0.3\text{--}0.55) \times (0.3\text{--}0.55) \times (1.8\text{--}3.4)$ m
Placement range x	[1.5, 5.0] m ahead of robot
Placement range y	$[-2.5, 2.5]$ m lateral
Minimum obstacle separation	1.25 m

Training Protocol

For PPO we adopt a two-stage training strategy:

1. **Stage 1 Empty world pre training.** Train for up to 50,000 iterations in $\mathcal{E}$ until the policy achieves stable forward locomotion (mean episode length ≈ 20 s, timeout rate $> 99\%$).
2. **Stage 2 Obstacle world fine tuning.** Initialise from the Stage 1 checkpoint and continue training in $\mathcal{W}$ with the obstacle reward terms and collision termination enabled.

For TD3 we train directly in each world from scratch due to the off cheppology nature of the algorithm.

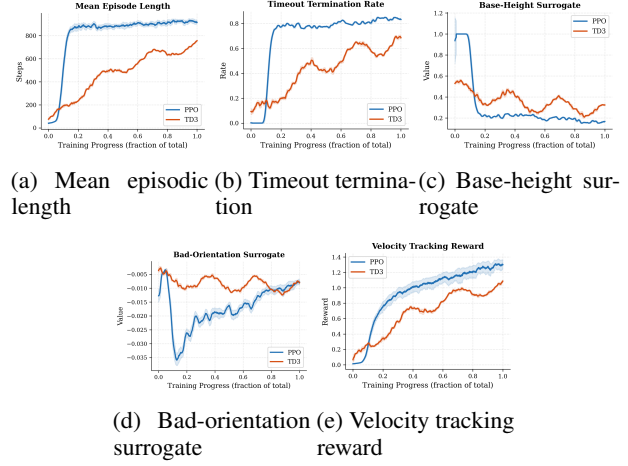


Figure 3: TD3 vs PPO in the empty-world no-LiDAR ablation. Curves show dense training traces with rolling mean and one-standard-deviation bands. This setting isolates locomotion stability and command following without obstacle contacts.

Results

We report results across six ablation conditions. For each condition we record the mean episode return (MER), mean episode length (MEL), and the collision rate (CR) the fraction of episodes that terminate due to obstacle contact where applicable. Training statistics are reported as mean $\pm$ std over the final 10% of training iterations; eval statistics are computed over 500 held out test episodes.

Ablation I: PPO vs. TD3 Empty World, No LiDAR

Algorithm	MEL (s) $\uparrow$	Timeout $\uparrow$	Vel. Track. $\uparrow$	Train Steps
PPO	18.4 ± 0.4	$83.8\% \pm 1.0\%$	1.281 ± 0.072	50,000 iters
TD3	14.0 ± 0.6	$64.8\% \pm 4.4\%$	1.004 ± 0.050	5M steps

PPO achieves a 31% longer mean episode length and 19-point higher timeout rate, confirming that ppo performs much better than TD3 and has fewer fall rate and have bigger episodic lengths.

Empty World No LiDAR. Figure 3 shows the empty world no LiDAR ablation. This setting provides a baseline for comparing algorithmic behavior when the robot only needs to maintain stable locomotion and track commanded velocity, without obstacle induced contacts.

Ablation II: PPO vs. TD3 : Obstacle World, No LiDAR

Algorithm	MEL (s) $\uparrow$	Timeout $\uparrow$	Vel. Track. $\uparrow$	Object Contact $\downarrow$
PPO	1.9 ± 0.3	$0.5\% \pm 0.3\%$	0.085 ± 0.010	-0.194 ± 0.021
TD3	2.1 ± 0.3	$12.5\% \pm 1.0\%$	0.092 ± 0.023	-0.337 ± 0.046

Without LiDAR, both algorithms fail to learn effective obstacle avoidance. Episode lengths collapse to under 2.1 s as the robot immediately contacts obstacles and falls. TD3

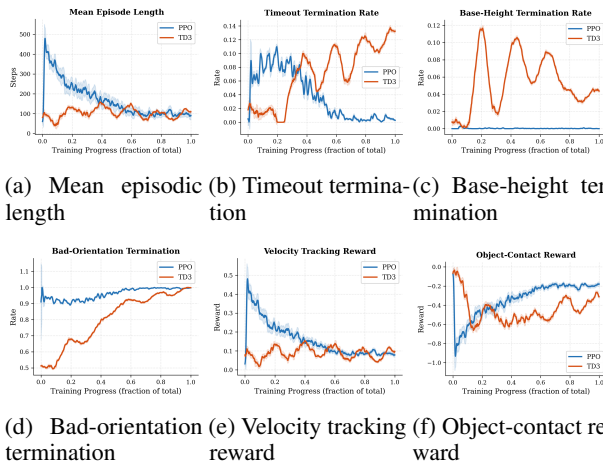


Figure 4: TD3 vs PPO in the obstacle world no LiDAR ablation. The additional object contact reward captures the obstacle collision penalty. This setting evaluates robustness to contact-rich interactions under partial observability.

incurs a 73% larger object contact penalty than PPO, suggesting more aggressive but less stable behaviour. The near zero timeout rates indicate that termination is overwhelmingly driven by obstacle collisions and subsequent falls rather than locomotion failure.

Ablation III & IV: Algorithm Comparison : With LiDAR

World	Algorithm	MER $\uparrow$	MEL (s) $\uparrow$	Timeout $\uparrow$	CR $\downarrow$
Empty	PPO + LiDAR	41.33 ± 0.67	19.9 ± 0.3	99.8%	N/A
Empty	TD3 + LiDAR	-2.12 ± 0.06	2.0 ± 0.1	0.0%	N/A
Obstacle	PPO + LiDAR	22.45 ± 1.01	19.4 ± 0.3	94.5%	4.2%
Obstacle	TD3 + LiDAR	-4.83 ± 0.06	1.0 ± 0.1	0.0%	0.0%

PPO with LiDAR achieves near maximal episode lengths in both worlds, with the obstacle world showing a modest 5.5-point drop in timeout rate due to unavoidable collisions in densely packed configurations. The mean episode return drops from 41.33 (empty) to 22.45 (obstacle), reflecting the additional object-hit and proximity penalties.

By contrast, TD3 with LiDAR fails to learn in both settings: episode lengths collapse to 2.0 s (empty) and 1.0 s (obstacle), with 100% bad-orientation termination in all episodes. The obstacle collision rate is 0% for TD3+LiDAR not because the robot avoids obstacles, but because it falls before reaching them identical to the behaviour of the empty-world transfer policy in Ablation V. This failure is attributable to the mismatch between TD3’s random policy rollout collection in warm start and the 1,720-dimensional LiDAR observation space: the replay buffer cannot accumulate sufficiently diverse experience to support stable Q-function learning at this input dimensionality. Figure 5 shows training curves for TD3 vs. PPO in the empty world with LiDAR, and Figure 6 shows PPO+LiDAR training in the obstacle world.

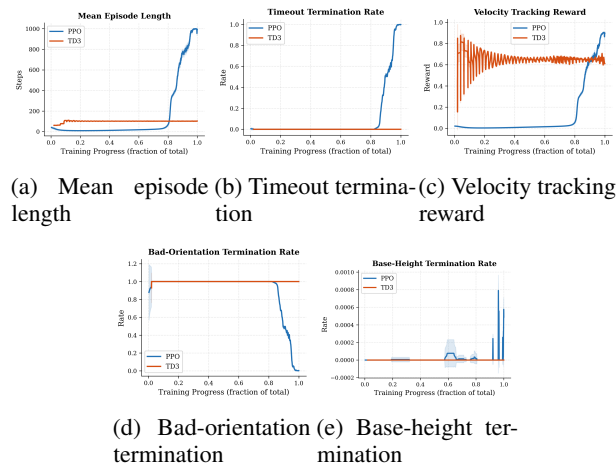


Figure 5: Ablation III: TD3 vs. PPO in the empty world *with* LiDAR observations. PPO converges to near-maximal episode length while TD3 immediately falls throughout all 5 M training steps, demonstrating that serial off-policy collection cannot learn from 1,720-dimensional LiDAR inputs.

Ablation V: PPO Transfer Empty World Policy in Obstacle World

Policy trained in	Evaluated in	MER $\uparrow$	MEL (s) $\uparrow$
Empty world (LiDAR)	Obstacle world	-13.87 ± 7.78	0.46 ± 0.28
Obstacle world (LiDAR)	Obstacle world	40.15 ± 20.72	16.0 ± 7.7

The transfer results reveal a striking gap: the obstacle-trained PPO and LiDAR policy survives $34\times$ longer (16.0 s vs. 0.46 s) and achieves a **53.9** point higher mean return than the empty world policy when both are evaluated in the dense obstacle world.

Figure 7 summarises the key metrics.

Results Discussion

PPO vs. TD3.

PPO vs. TD3. PPO consistently outperforms TD3 in the empty world, achieving a 31% higher mean episode length and a higher velocity-tracking reward (1.28 vs. 1.00). This result is notable because TD3 is often regarded as a strong choice for continuous control problems, yet under our setting PPO shows clearly better locomotion performance. In the obstacle world without LiDAR, the performance gap narrows substantially, as both algorithms fail to learn effective avoidance. However, TD3 incurs larger collision penalties, suggesting less stable interaction with obstacles. With LiDAR observations, the difference becomes qualitative rather than incremental: PPO learns strong obstacle aware locomotion, reaching a 94.5% timeout rate in the obstacle world, whereas TD3 fails to stand up reliably, with episode lengths of at most 2 s and a 100% fall rate. This is one of the main findings of our study: although TD3 is generally expected to perform well in continuous domains, PPO is markedly more effective in this locomotion setting, especially once high dimensional perception is introduced.

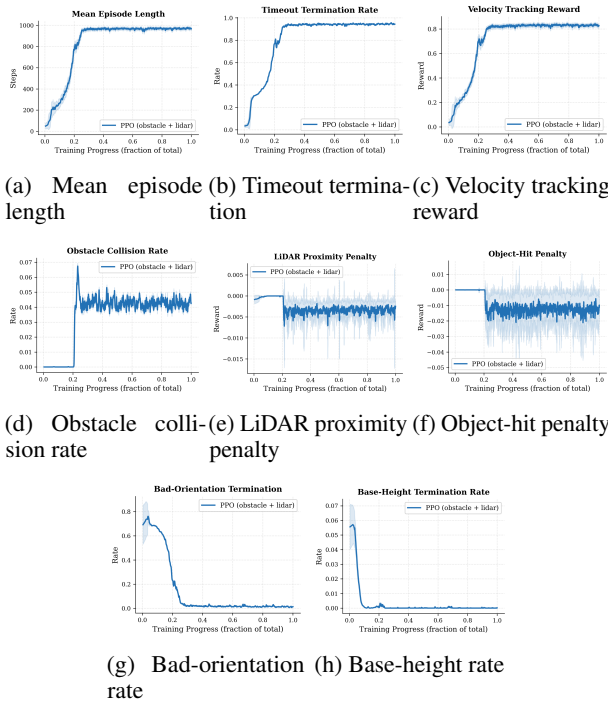


Figure 6: Ablation IV: PPO+LiDAR training in the dense obstacle world (fine-tuned from the empty-world checkpoint). The policy reaches a 94.5% timeout rate and a 4.2% collision rate, with the LiDAR proximity penalty driving anticipatory obstacle avoidance before physical contact. TD3+LiDAR results are omitted as the robot falls immediately throughout training (see Figure 5).

Effect of LiDAR. LiDAR provides a transformative benefit for PPO in the obstacle world. PPO with LiDAR achieves a 94.5% timeout rate, compared to near zero for the no LiDAR baseline, showing that anticipatory avoidance learned through the `lidar_proximity` reward allows the policy to avoid obstacles before contact. In the empty world, LiDAR provides only marginal benefit for PPO: PPO with LiDAR achieves 19.9 s mean episode length compared to 18.4 s without LiDAR. This suggests that, in the absence of obstacles, the additional high dimensional input slightly increases learning difficulty without contributing meaningful information.

For TD3, LiDAR does not provide any benefit. In fact, TD3 with LiDAR performs worse than TD3 without LiDAR, with episode lengths of 2.0 s in the empty world and 1.0 s in the obstacle world, compared to 14.1 s and 2.1 s without LiDAR. Overall, these results indicate that adding LiDAR observations changes the learning problem in a way that PPO handles far more effectively than TD3 in our setup.

Transfer. Two-stage training (pre-train on empty, fine-tune on obstacles) is critical: the directly fine-tuned obstacle policy achieves $34\times$ longer episodes than the zero-shot transfer baseline. The transfer policy’s failure is locomotion destabilisation rather than obstacle collision, indicating that obstacle aware reward shaping is necessary even for a policy that has

already mastered walking.

Future Work

- **Sim-to-real transfer.** Deploying the best obstacle aware policy on the physical Unitree G1 using domain randomisation over physics parameters and sensor noise.
- **Curriculum learning.** Automating the progression from empty to dense-obstacle environments via adaptive difficulty, e.g., PAIRED [Dennis et al. 2020] or automatic curriculum generation based on episode return.
- **Multi modal fusion.** Combining LiDAR with onboard camera data for richer scene understanding, potentially using a learned encoder rather than raw range values.
- **Dynamic obstacles.** Extending the obstacle world to include more moving objects (other robots, pedestrians) and evaluating whether the LiDAR policy generalises without retraining.
- **Whole body obstacle avoidance.** Extending the contact sensor to all body links (not just the torso) and adding per-body proximity penalties to encourage avoidance with arms, head, and legs.

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Reward Function Definitions

Velocity Tracking

$$r_{\text{vel}} = \exp\left(-\frac{\|\mathbf{v}_{xy}^{\text{cmd}} - \mathbf{v}_{xy}\|^2}{\sigma^2}\right), \quad \sigma^2 = 0.25 \quad (5)$$

Gait Reward

$$r_{\text{gait}} = \sum_{i=1}^2 \mathbf{1}[\text{stance}_i(t) = \text{contact}_i(t)] \quad (6)$$

where

$$\text{stance}_i(t) = \mathbf{1}[\phi_i(t) < 0.55], \quad (7)$$

$$\phi_i(t) = \left(\frac{t}{T_g} + \Delta\phi_i\right) \bmod 1, \quad (8)$$

with gait period

$$T_g = 0.8 \text{ s}, \quad \Delta\phi_1 = 0, \quad \Delta\phi_2 = 0.5. \quad (9)$$

LiDAR Proximity Penalty

$$r_{\text{lidar}} = -\sum_{k=1}^K \frac{\max(0, d_{\text{safe}} - d_k)}{d_{\text{safe}}}, \quad (10)$$

$$d_{\text{safe}} = 0.8 \text{ m}, \quad K = 248$$

Observation Space Details

Table 1: Observation-space dimensions for actor and critic inputs.

Observation group	Modality	Per-step dim.	With history ($T = 5$)
Policy — no LiDAR	Proprioception	96	480
Policy — with LiDAR	Proprioception + LiDAR	344	1,720
Critic — no LiDAR	Proprioception+	99	495
Critic — with LiDAR	Proprioception+ + LiDAR	347	1,735

The critic’s privileged observation augments the policy observation with the ground-truth base linear velocity in $\mathbb{R}^3$. This additional signal is used only during training to reduce value-estimation variance, while the actor remains blind to it at deployment.

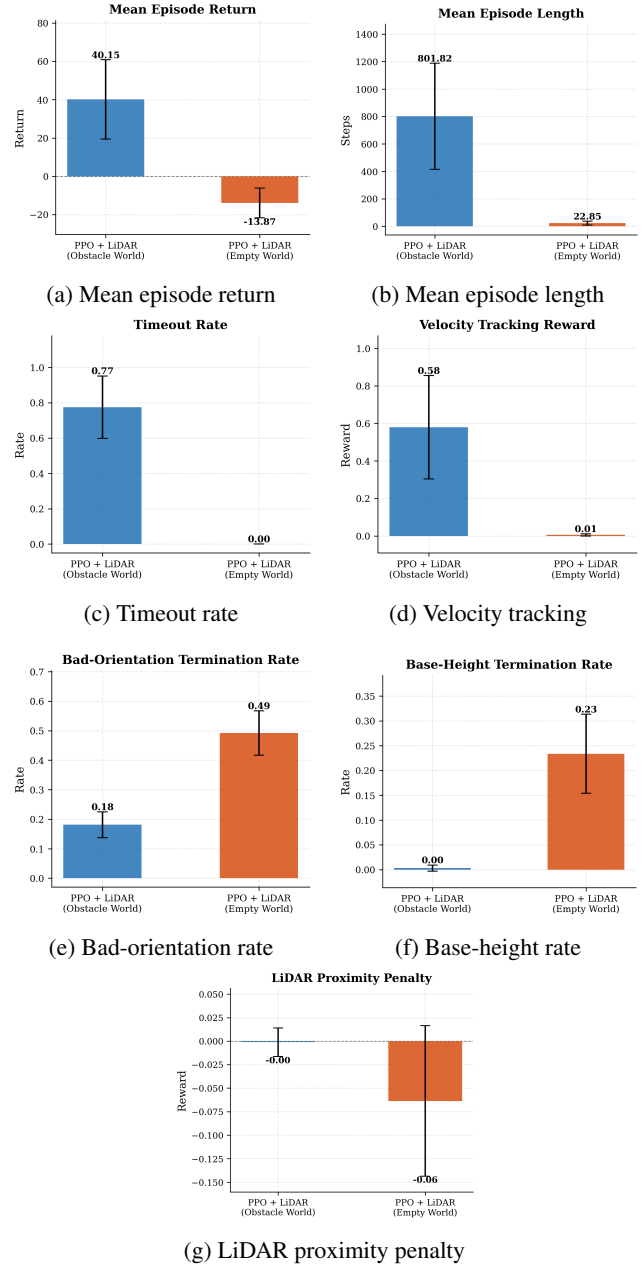


Figure 7: Ablation V: obstacle-trained PPO+LiDAR policy (blue) vs. empty-world PPO+LiDAR policy transferred to the dense obstacle world (orange), evaluated over 500 episodes each. Row 1: performance metrics (return, episode length, timeout rate, velocity tracking). Row 2: failure and avoidance metrics (bad orientation, base height, collision rate, LiDAR proximity penalty). The obstacle-trained policy dominates across all eight metrics.